```
-- queries.sql
-- Nassau Candy Distributor, shipping route
efficiency
-- Standard SQL, tested as MySQL
compatible. Run against the cleaned
-- table produced by data_cleaning.py,
loaded as a table called `shipments`
-- with columns: Row_ID, Order_ID,
Order_Date, Ship_Date, Ship_Mode,
-- Customer_ID, Country_Region, City,
State_Province, Postal_Code,
-- Division, Region, Product_ID,
Product_Name, Sales, Units, Gross_Profit,
-- Cost, Shipping_Lead_Time_Days, Factory,
Route, Order_Month


-- Query 1: overall KPIs
-- How big is the shipping operation and
how efficient is it overall,
-- total volume, total transactions,
average lead time.
SELECT
    COUNT(*)
AS total_shipments,
    COUNT(DISTINCT Order_ID)
AS total_orders,
    ROUND(SUM(Sales), 2)
AS total_sales,
    ROUND(AVG(Sales), 2)
```

```
AS avg_order_value,
    ROUND(AVG(Shipping_Lead_Time_Days), 2)
AS avg_lead_time_days
FROM shipments;


-- Query 2: trend over time
-- Is shipping performance improving or
getting worse month to month,
-- and are certain months consistently
slower.
SELECT
    Order_Month,
    COUNT(*)
AS shipment_count,
    ROUND(SUM(Sales), 2)
AS monthly_sales,
    ROUND(AVG(Shipping_Lead_Time_Days), 2)
AS avg_lead_time_days
FROM shipments
GROUP BY Order_Month
ORDER BY Order_Month;


-- Query 3: breakdown by region and ship
mode
-- Which ship modes and regions carry the
most volume, and how does
-- their average lead time compare.
SELECT
    Region,
    Ship_Mode,
    COUNT(*)
AS shipment_count,
```

```
    ROUND(AVG(Shipping_Lead_Time_Days), 2)
AS avg_lead_time_days,
    ROUND(SUM(Sales), 2)
AS total_sales
FROM shipments
GROUP BY Region, Ship_Mode
ORDER BY shipment_count DESC;


-- Query 4: route efficiency ranking
-- Which factory to customer routes are
fastest and slowest, ranked by
-- average lead time, limited to routes
with at least 5 shipments so the
-- ranking isn't skewed by one-off
outliers. Uses a CTE plus RANK().
WITH route_stats AS (
    SELECT
        Route,
        Factory,
        State_Province,

COUNT(*)                                AS
shipment_count,
        ROUND(AVG(Shipping_Lead_Time_Days),
2) AS avg_lead_time_days
    FROM shipments
    GROUP BY Route, Factory, State_Province
    HAVING COUNT(*) >= 5
)
SELECT
    Route,
    Factory,
    State_Province,
```

```
    shipment_count,
    avg_lead_time_days,
    RANK() OVER (ORDER BY
avg_lead_time_days ASC)  AS
efficiency_rank_fastest,
    RANK() OVER (ORDER BY
avg_lead_time_days DESC) AS
efficiency_rank_slowest
FROM route_stats
ORDER BY avg_lead_time_days ASC;


-- Query 5: geographic bottlenecks
-- Which states have both high shipment
volume and above average lead
-- time at the same time, since that's
where logistics investment would
-- probably matter most.
SELECT
    State_Province,
    COUNT(*)
AS shipment_count,
    ROUND(AVG(Shipping_Lead_Time_Days), 2)
AS avg_lead_time_days
FROM shipments
GROUP BY State_Province
HAVING
    COUNT(*) > (SELECT AVG(state_volume)
FROM (
        SELECT COUNT(*) AS state_volume
        FROM shipments
        GROUP BY State_Province
    ) AS volume_by_state)
    AND AVG(Shipping_Lead_Time_Days) >
```

```
(SELECT AVG(Shipping_Lead_Time_Days) FROM
shipments)
ORDER BY avg_lead_time_days DESC;
```